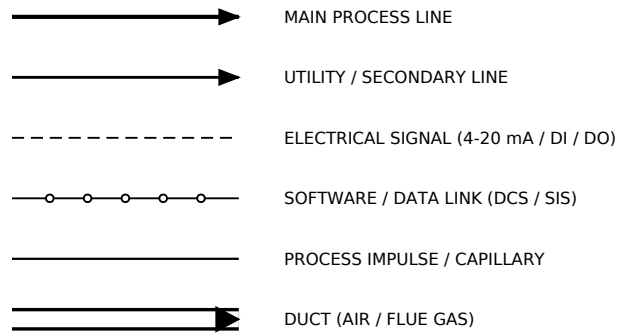
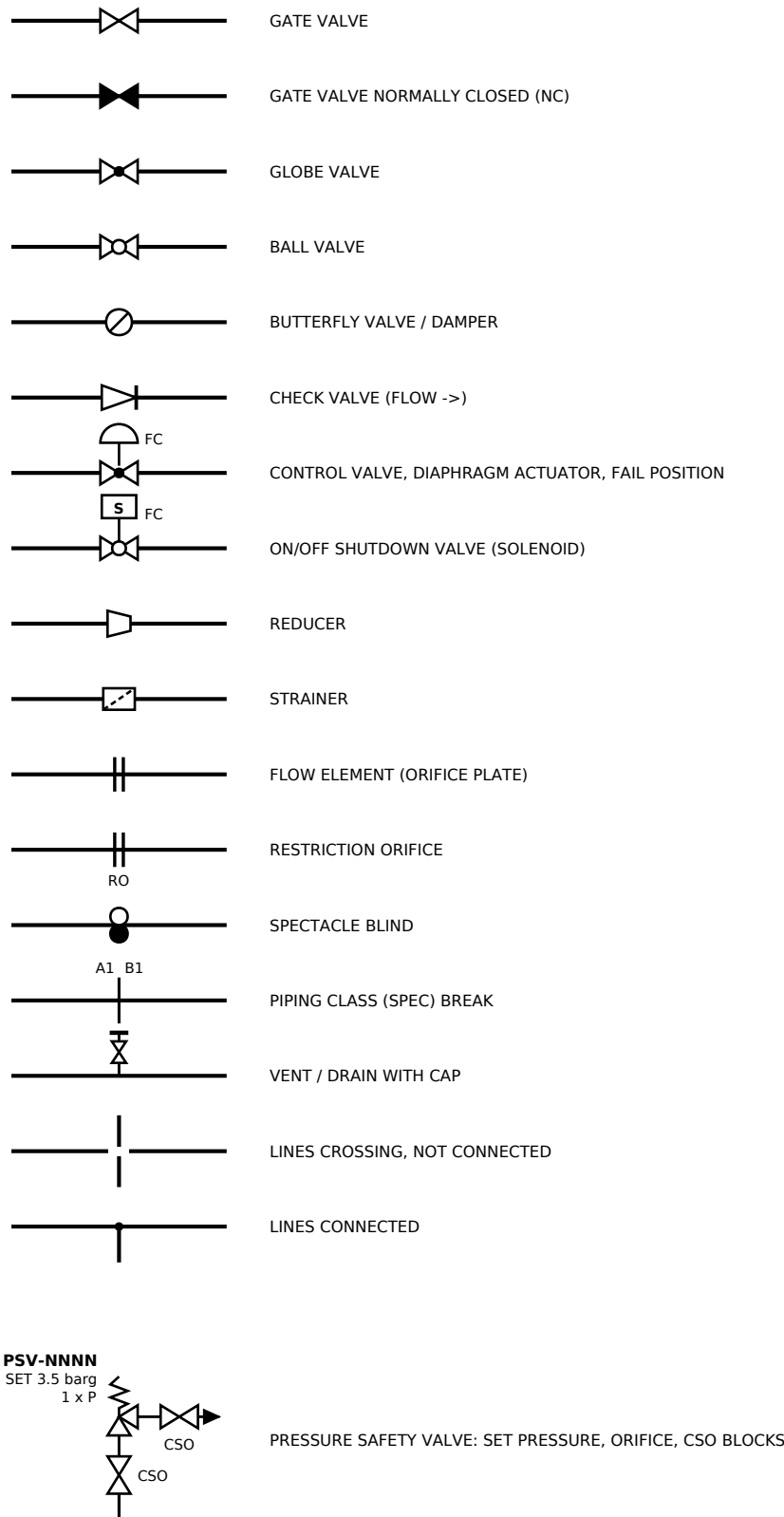


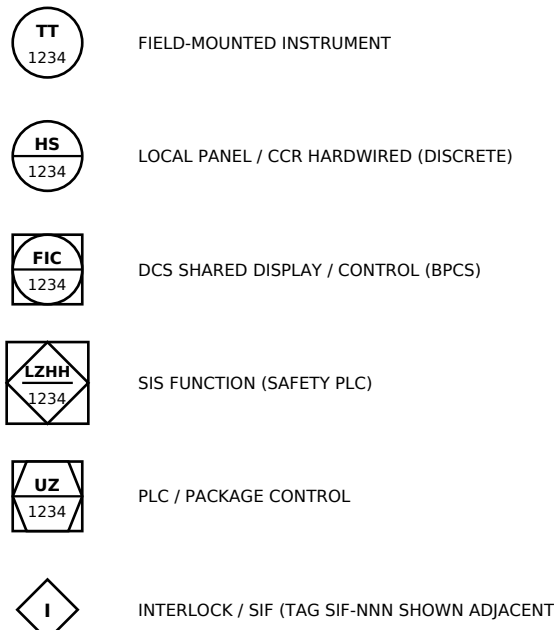
LINE SYMBOLS



VALVES & PIPING COMPONENTS



INSTRUMENT SYMBOLS (ISA-5.1)



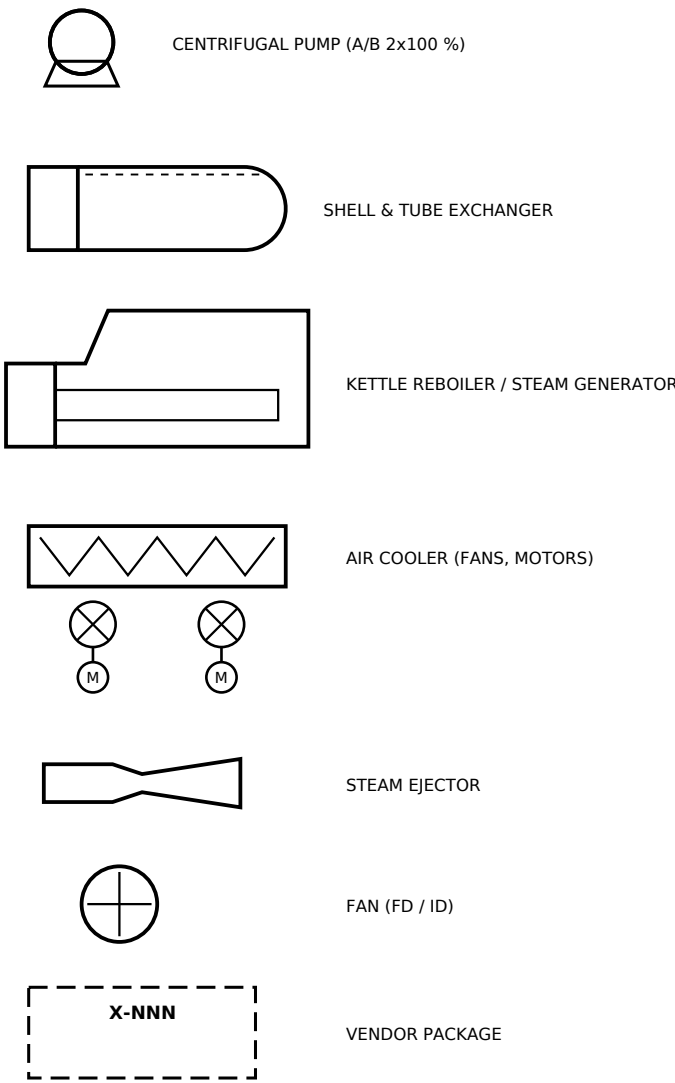
INSTRUMENT IDENTIFICATION LETTERS

A ANALYSIS (O₂, pH) B BURNER / FLAME
F FLOW H HAND
L LEVEL P PRESSURE
T TEMPERATURE X/Z ON-OFF / POSITION
PD / TD DIFFERENTIAL FF FLOW RATIO
Succeeding: E element, G glass, I indicate, C control,
T transmit, V valve, Y relay/compute, S switch, Z (2x4) SIS,
AH/AH/ALL alarms; A/B/C suffix = redundant SIS

ABBREVIATIONS

FC / FO / FL FAIL CLOSED / OPEN / LAST
CSO / CSC CAR-SEALED OPEN / CLOSED
NC / NO NORMALLY CLOSED / OPEN
NMF NORMALLY NO FLOW
TPA / MPA / BPA TOP / MID / BOTTOM PUMPAROUND
SS STRIPPING STEAM
DBB DOUBLE BLOCK & BLEED
ROSOV / EIV REMOTE OPERATED SHUT-OFF VALVE
COI / COT COIL OUTLET / INLET TEMP.
RO RESTRICTION ORIFICE
FA FLAME ARRESTOR
OSBL OUTSIDE BATTERY LIMITS
BMS BURNER MANAGEMENT SYSTEM
2oo3 TWO-OUT-OF-THREE VOTING

EQUIPMENT SYMBOLS



LINE NUMBER

12"-P-100-002-B1-H

NPS (in) - FLUID - AREA - SEQUENCE - PIPING CLASS - INSULATION
AREA: 100 CDU, 200 VDU, 900 UNIT UTILITIES

FLUID CODES

P Process hydrocarbon
PG Process gas / vapour
SW Sour water
WW Wash water / brine
LS LP steam
HS HP steam
CD Condensate
BFW Boiler feed water

CWS Cooling water supply
CWR Cooling water return
FG Fuel gas
FL Flare
BD Blowdown / drain
IA Instrument air
N Nitrogen
CH Chemical

INSULATION CODES

H HEAT CONSERVATION
P PERSONNEL PROTECTION
ST STEAM TRACED + INSULATED
C COLD
N NONE

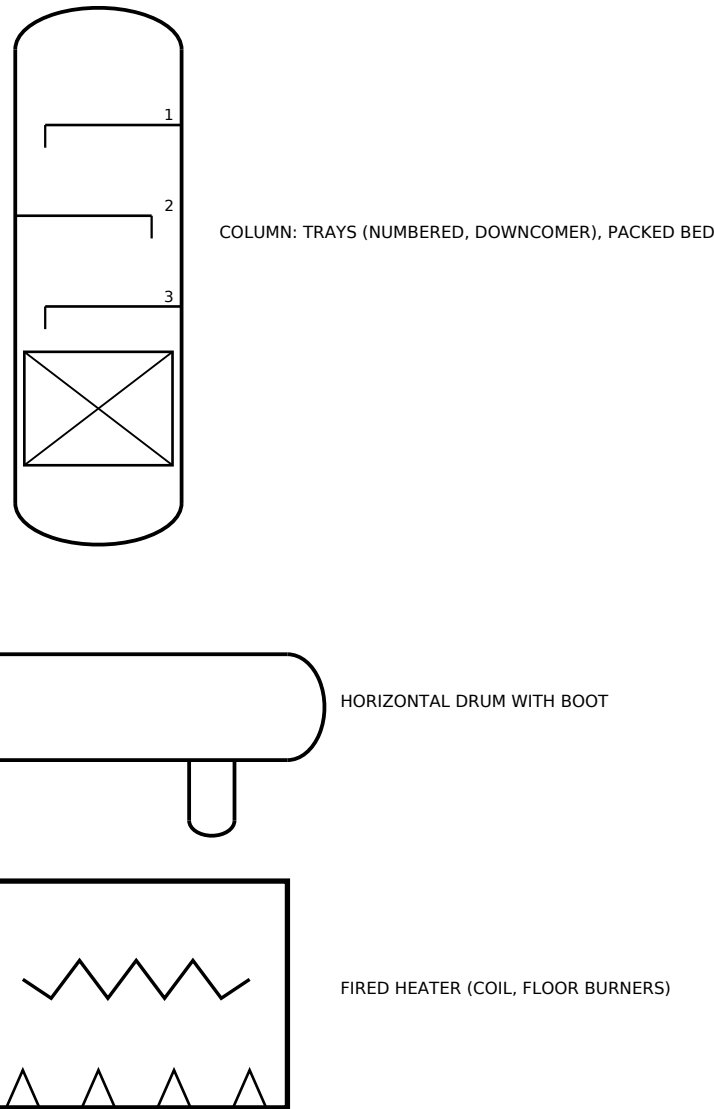
PIPING CLASSES (SUMMARY, SEE CFU-000-PI-SPC-001)

A1 150# Carbon steel ASTM A106 Gr.B
A2 150# Carbon steel, HIC-resistant (NACE MR0103)
B1 300# Carbon steel ASTM A106 Gr.B
B2 300# SC-1/2Mo ASTM A335 P5
B3 300# 9Cr-1Mo ASTM A335 P9
C1 600# Carbon steel, killed ASTM A106 Gr.B
A3 300# Carbon steel, HIC-resistant (NACE MR0103)
S1 150# Carbon steel ASTM A106 Gr.B
S2 600# 1.25Cr-1/2Mo ASTM A335 P11
S3 300# Carbon steel ASTM A106 Gr.B
F1 150# Carbon steel, killed, impact-tested (A33)
U1 150# Carbon steel (cement-lined for CW >= 12")
U2 150# SS304 (A) / galvanised CS (N2)
DRAWING INDEX-335 P5

Hydrocarbon <= 230 C, general
Sour water, OH vapour/condensate, wash water/brine
Hydrocarbon <= 260 C, pump discharges
Sulfidic hydrocarbon 260-400 C (incl crude, AR, BPA, HvGO, VR)
Heater outlets / transfer lines (H-101, H-201) and > 400 C
LPG / stabiliser, BFW > 300#
Wash water / brine / sour water above 150# rating (desalter pr
LP / HP steam, condensate
HP steam 41 barg / 400 C
MP steam 10.3 barg / 250-290 C design (exceeds S1)
Flare / relief headers, -29 to 350 C
Cooling water, utility water, LP BFW
Instrument air, nitrogen
Vacuum vapour lines (C-201 OH); transfer line > 400 C -> B3

LEGEND, SYMBOLS & ABBREVIATIONS
CRUDE CHARGE PUMPS & COLD PREHEAT TRAIN
ELECTROSTATIC DESALTERS D-101A/B & BOOSTER PUMPS
HOT PREHEAT TRAIN E-106 TO E-111 & E-113
ATMOSPHERIC HEATER H-101 - PROCESS COILS
ATMOSPHERIC HEATER H-101 - FIRING & AIR SYSTEM
ATMOSPHERIC COLUMN C-101 - FLASH ZONE & BOTTOMS
ATMOSPHERIC COLUMN C-101 - UPPER SECTION & PUMPAROUNDS
SIDE STRIPPERS C-102/103/104 & PRODUCT RUNDOWN
C-101 OVERHEAD SYSTEM & REFLUX DRUM D-102
NAPHTHA STABILISER C-105
NAPHTHA SPLITTER C-106
VACUUM HEATER H-201
VACUUM COLUMN C-201
VACUUM PUMPAROUNDS & PRODUCTS
VACUUM EJECTOR SYSTEM & HOTWELL D-201
FUEL GAS, FLARE KO & UTILITY HEADERS

EQUIPMENT DATA BOX



NOTES

- Legend applies to all P&IDs CFU-xxx-PR-PID-001 to -016.
- F&G devices (GD gas / H₂S, BE flame, HS manual call points) are listed in instrument index CFU-000-IC-IDX-001; F&G layout drawing by others.
- DCS computing blocks (FY / PY / FFY) per control scheme drawings CFU-xxx-IC-CSD; listed in index.

REV	DATE	DESCRIPTION		
A	2026-10-02	Issued for review (FEED)		
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT				
PIPING & INSTRUMENT DIAGRAM LEGEND, SYMBOLS & ABBREVIATIONS				
DISCIPLINE		STAGE	SCALE	SHEET
PROCESS		FEED	NTS	1 OF 17
DRAWN		CHECKED	APPROVED	DATE
CLAUDE CODE		-	-	2026-10-02
DRAWING NUMBER				REV
CFU-000-PR-PID-000				A